

# **Companion to Bird & de Moor, Algebra of Programming**

## §0. Laws from Relation Algebra

### §0.1. Adjunctions

(0.1a)

|    | $F \dashv G$                                            | F's type                                                              | monad $FG$                            | $1 \sqsubseteq FG$                                                             | $GF \sqsubseteq 1$                                                             | F preserves $\cup$                                                | G preserves $\cap$                                                                     | $FGF=F$                                                         | $GFG=G$                                                           |
|----|---------------------------------------------------------|-----------------------------------------------------------------------|---------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------|-------------------------------------------------------------------|----------------------------------------------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------|
| 1  | $\triangleleft \dashv \triangleright$                   | $A \rightarrow A \otimes A$                                           | $\triangleleft \triangleright = 1$    | $1 \sqsubseteq \triangleleft \triangleright$                                   | $\triangleright \triangleleft \sqsubseteq 1$                                   | $(R \cup S) \triangleleft = R \triangleleft \cup S \triangleleft$ | $(R \cap S) \triangleright = R \triangleright \cap S \triangleright$                   | $\triangleleft \triangleright \triangleleft = \triangleleft$    | $\triangleright \triangleleft \triangleright = \triangleright$    |
| 2  | $\multimap \dashv \multimap$                            | $A \rightarrow I$                                                     | $\multimap \multimap = T$             | $1 \sqsubseteq \multimap \multimap$                                            | $\multimap \multimap \sqsubseteq 1$                                            | $(R \cup S) \multimap = R \multimap \cup S \multimap$             | $(R \cap S) \multimap = R \multimap \cap S \multimap$                                  | $\multimap \multimap \multimap = \multimap$                     | $\multimap \multimap \multimap = \multimap$                       |
| 3  | $\circ \dashv \circ$                                    | $(A \rightarrow B) \rightarrow (B \rightarrow A)$                     | $R^{\circ \circ} = R$                 | $R^{\circ \circ} = R$                                                          | $R^{\circ \circ} = R$                                                          | $(R \cup S)^{\circ} = S^{\circ} \cup R^{\circ}$                   | $(R \cap S)^{\circ} = R^{\circ} \cap S^{\circ}$                                        | $R^{\circ \circ} = R$                                           | $R^{\circ \circ} = R$                                             |
| 4  | $\multimap \triangleleft \triangleright \multimap$      | $(X \otimes A \rightarrow Y) \rightarrow (X \rightarrow Y \otimes A)$ | $1$                                   | $(\multimap \triangleleft \otimes 1) (1 \otimes \triangleright \multimap) = 1$ | $(1 \otimes \multimap \triangleleft) (\triangleright \multimap \otimes 1) = 1$ | $=$                                                               | $=$                                                                                    | $=$                                                             | $=$                                                               |
| 5  | $\triangleright \multimap \triangleleft \triangleright$ | $(X \rightarrow Y \otimes A) \rightarrow (X \otimes A \rightarrow Y)$ | $1$                                   | $(1 \otimes \multimap \triangleleft) (\triangleright \multimap \otimes 1) = 1$ | $(\multimap \triangleleft \otimes 1) (1 \otimes \triangleright \multimap) = 1$ | $=$                                                               | $=$                                                                                    | $=$                                                             | $=$                                                               |
| 6  | $\Delta \dashv \cap$                                    | $(A \rightarrow B) \rightarrow (A \rightarrow B)^2$                   | $R \cap R = R$                        | $R \cap R = R$                                                                 | $R \cap S \sqsubseteq R$                                                       | $\Delta(R \cup S) = \Delta R \cup \Delta S$                       | $(R \cap T) \cap (S \cap U) = (R \cap S) \cap (T \cap U)$                              | $R \cap R = R$                                                  | $R \cap R = R$                                                    |
| 7  | $\cup \dashv \Delta$                                    | $(A \rightarrow B)^2 \rightarrow (A \rightarrow B)$                   | $(R, S) \mapsto (R \cup S, R \cup S)$ | $R \sqsubseteq R \cup S$                                                       | $R \cup R = R$                                                                 | $(R \cup T) \cup (S \cup U) = (R \cup S) \cup (T \cup U)$         | $\Delta(R \cap S) = \Delta R \cap \Delta S$                                            | $R \cup R = R$                                                  | $R \cup R = R$                                                    |
| 8  | $\emptyset \dashv !$                                    | $\{*\} \rightarrow (A \rightarrow B)$                                 |                                       |                                                                                | $\emptyset \sqsubseteq R$                                                      |                                                                   |                                                                                        |                                                                 |                                                                   |
| 9  | $\cdot S \dashv / S$                                    | $(A \rightarrow B) \rightarrow (A \rightarrow C)$                     | $S/S$                                 | $R \sqsubseteq (RS)/S$                                                         | $(T/S)S \sqsubseteq T$                                                         | $(S \cup T)R = (SR) \cup (TR)$                                    | $(R \cap R') / S = (R / S) \cap (R' / S)$                                              | $((RS)/S)S = RS$                                                | $((T/S)S)/S = T/S$                                                |
| 10 | $S \cdot \dashv S \setminus$                            | $(B \rightarrow C) \rightarrow (A \rightarrow C)$                     | $S \setminus S$                       | $R \sqsubseteq S \setminus (SR)$                                               | $S(S \setminus T) \sqsubseteq T$                                               | $R(S \cup T) = (RS) \cup (RT)$                                    | $S \setminus (R \cap R') = (S \setminus R) \cap (S \setminus R')$                      | $S(S \setminus (SR)) = SR$                                      | $S \setminus (S(S \setminus T)) = S \setminus T$                  |
| 11 | $R \cap \dashv R \Rightarrow$                           | $(A \rightarrow B) \rightarrow (A \rightarrow B)$                     | $X \mapsto R \Rightarrow (X \cap R)$  | $X \sqsubseteq R \Rightarrow (X \cap R)$                                       | $R \cap (R \Rightarrow Y) \sqsubseteq Y$                                       | $R \cap (X \cup Y) = (R \cap X) \cup (R \cap Y)$                  | $R \Rightarrow S \cap T = (R \Rightarrow S) \cap (R \Rightarrow T)$                    | $R \cap (R \Rightarrow (X \cap R)) = X \cap R$                  | $R \Rightarrow (R \cap (R \Rightarrow Y)) = R \Rightarrow Y$      |
| 12 | $\mathcal{D} \dashv \cdot T$                            | $(A \rightarrow B) \rightarrow \text{Cor } A$                         | $R \mapsto (\mathcal{D}R)T$           | $R \sqsubseteq (\mathcal{D}R)T$                                                | $\mathcal{D}(AT) \sqsubseteq A$                                                | $\mathcal{D}(R \cup S) = \mathcal{D}R \cup \mathcal{D}S$          | $(A \cap B)T = AT \cap BT$                                                             | $\mathcal{D}((\mathcal{D}R)T) = \mathcal{D}R$                   | $(\mathcal{D}(AT))T = AT$                                         |
| 13 | $\mathcal{R} \dashv T \cdot$                            | $(A \rightarrow B) \rightarrow \text{Cor } B$                         | $R \mapsto T(\mathcal{R}R)$           | $R \sqsubseteq T(\mathcal{R}R)$                                                | $\mathcal{R}(TA) \sqsubseteq A$                                                | $\mathcal{R}(R \cup S) = \mathcal{R}R \cup \mathcal{R}S$          | $T(A \cap B) = TA \cap TB$                                                             | $\mathcal{R}(T(\mathcal{R}R)) = \mathcal{R}R$                   | $T(\mathcal{R}(TA)) = TA$                                         |
| 14 | $\cdot f \dashv \cdot f^{\circ}$                        | $(A \rightarrow B) \rightarrow (A \rightarrow C)$                     | $f f^{\circ}$                         | $\text{Map}(f) \Rightarrow 1 \sqsubseteq f f^{\circ}$                          | $f^{\circ} f \sqsubseteq 1$                                                    | $(S \cup T)R = (SR) \cup (TR)$                                    | $\text{Simple}(g) \Rightarrow (X \cap Y) g^{\circ} = (X g^{\circ}) \cap (Y g^{\circ})$ | $f f^{\circ} f = f$                                             | $f^{\circ} f f^{\circ} = f^{\circ}$                               |
| 15 | $f^{\circ} \cdot \dashv f \cdot$                        | $(A \rightarrow C) \rightarrow (B \rightarrow C)$                     | $f f^{\circ}$                         | $\text{Map}(f) \Rightarrow 1 \sqsubseteq f f^{\circ}$                          | $f^{\circ} f \sqsubseteq 1$                                                    | $R(S \cup T) = (RS) \cup (RT)$                                    | $f(X \cap Y) = fX \cap fY$                                                             | $f f^{\circ} f = f$                                             | $f^{\circ} f f^{\circ} = f^{\circ}$                               |
| 16 | $i \dashv E$                                            | $\text{Map} \multimap \text{Rel}$                                     | $E$                                   | $\frac{1}{\exists} : A \rightarrow PA$                                         | $\exists : PB \rightarrow B$                                                   | —                                                                 | —                                                                                      | $\frac{\exists}{\exists} = 1$                                   | $\frac{R}{\exists} \exists = R$                                   |
| 17 | $\cdot \exists \dashv \frac{\exists}{\exists}$          | $\text{Map}(A, PB) \rightarrow (A \rightarrow B)$                     | $1$                                   | $\frac{f \exists}{\exists} = f$                                                | $\frac{R}{\exists} \exists = R$                                                | —                                                                 | —                                                                                      | $\frac{\exists}{\exists} \frac{f \exists}{\exists} = f \exists$ | $\frac{R}{\exists} \frac{f \exists}{\exists} \exists = f \exists$ |
| 18 | $\frac{\exists}{\exists} \dashv \cdot \exists$          | $(A \rightarrow B) \rightarrow \text{Map}(A, PB)$                     | $1$                                   | $\frac{R}{\exists} \exists = R$                                                | $\frac{f \exists}{\exists} = f$                                                | —                                                                 | —                                                                                      | $\frac{\exists}{\exists} \frac{R}{\exists} = \frac{R}{\exists}$ | $\frac{f \exists}{\exists} \frac{R}{\exists} \exists = f \exists$ |

## §0.2. Composing adjunctions

(0.2a)

|   | $\cdot T$             | $T \cdot$                                               | $\cdot g$                                                        | $g^\circ \cdot$                                   | $Tn$                                                      | $^\circ$                                                                                                                     | $\circ \triangleleft$                                                                                                        | $\Delta$                                                                         |
|---|-----------------------|---------------------------------------------------------|------------------------------------------------------------------|---------------------------------------------------|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| 1 | $\bullet S$           | $R / (S_1 \ S_2)$<br>$= R / S_2 / S_1$                  | $S \setminus (R / W)$<br>$= (S \setminus R) / W$                 | $R / (Sg) = (Rg^\circ) / S$                       | $\text{Map}(f) \Rightarrow f$<br>$(R / S) = (f R) / S$    | $(Y/S)^\circ = S^\circ \setminus (Y^\circ)$                                                                                  | $(RS)^\wedge = R^\wedge (S \otimes 1)$                                                                                       | $(R \cap R') / S = (R / S) \cap (R' / S)$                                        |
| 2 | $S \bullet$           | $S \setminus (R / W)$<br>$= (S \setminus R) / W$        | $(S_1 \ S_2) \setminus R$<br>$= S_2 \setminus (S_1 \setminus R)$ | $S \setminus (Rg^\circ) = (S \setminus R)g^\circ$ | $(g^\circ S) \setminus R = S \setminus (gR)$              | $(S \setminus Y)^\circ = Y^\circ / S^\circ$                                                                                  | $((S \otimes 1)R)^\wedge = SR^\wedge$                                                                                        | $S \setminus (R \cap R') = (S \setminus R) \cap (S \setminus R')$                |
| 3 | $\bullet f$           | $\text{Map}(f) \Rightarrow R / (f S) = (R / S) f^\circ$ | $T \setminus (Rf^\circ) = (T \setminus R) f^\circ$               | $(R S)^\circ = S^\circ R^\circ$                   |                                                           | $(R S)^\circ = S^\circ R^\circ$                                                                                              |                                                                                                                              | $\text{Simple}(g) \Rightarrow (X \cap Y) g^\circ = (X g^\circ) \cap (Y g^\circ)$ |
| 4 | $f^\circ \bullet$     | $\text{Map}(f) \Rightarrow f (R / S) = (f R) / S$       | $(Tf^\circ) \setminus R = f(T \setminus R)$                      |                                                   | $(R S)^\circ = S^\circ R^\circ$                           | $(R S)^\circ = S^\circ R^\circ$                                                                                              |                                                                                                                              | $f(T_1 \cap T_2) = fT_1 \cap fT_2$                                               |
| 5 | $Rn$                  | —                                                       | —                                                                | —                                                 | $R \Rightarrow S \Rightarrow T = R \cap S \Rightarrow T$  | $(R \Rightarrow Y)^\circ = R^\circ \Rightarrow (Y^\circ)$                                                                    | —                                                                                                                            | $R \Rightarrow S \cap T = (R \Rightarrow S) \cap (R \Rightarrow T)$              |
| 6 | $^\circ$              | $(Y/T)^\circ = T^\circ \setminus (Y^\circ)$             | $(T \setminus Y)^\circ = Y^\circ / T^\circ$                      | $(R S)^\circ = S^\circ R^\circ$                   | $(T \Rightarrow Y)^\circ = T^\circ \Rightarrow (Y^\circ)$ | $R^\circ{}^\circ = R$                                                                                                        | $(R^\wedge)^\circ = (R^\circ \otimes 1)$<br>$(1 \otimes \triangleright \circ)$<br>both bends: the $^\circ$ section's display | $(T_1 \cap T_2)^\circ = T_1^\circ \cap T_2^\circ$                                |
| 7 | $\circ \triangleleft$ | $(RT)^\wedge = R^\wedge (T \otimes 1)$                  | $((T \otimes 1)R)^\wedge = TR^\wedge$                            | —                                                 | —                                                         | $(R^\wedge)^\circ = (R^\circ \otimes 1)$<br>$(1 \otimes \triangleright \circ)$<br>both bends: the $^\circ$ section's display | —                                                                                                                            | —                                                                                |
| 8 | $u$                   | $(S \cup T) R = (S R) \cup (T R)$                       | $R (S \cup T) = (R S) \cup (R T)$                                | $(S \cup T) R = (S R) \cup (T R)$                 | $R (S \cup T) = (R S) \cup (R T)$                         | $R \cap (S \cup T) = R \cap S \cup R \cap T$                                                                                 | $(R \cup S)^\circ = S^\circ \cup R^\circ$                                                                                    |                                                                                  |
| 9 | $\emptyset$           | $\emptyset R = \emptyset$                               | $R \emptyset = \emptyset$                                        | $\emptyset R = \emptyset$                         | $R \emptyset = \emptyset$                                 | $R \cap \emptyset = \emptyset$                                                                                               | $\emptyset^\circ = \emptyset$                                                                                                |                                                                                  |

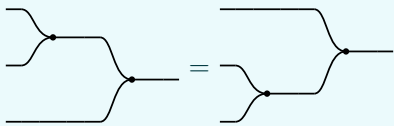
## §0.3. Adjoint triples

(0.3a)

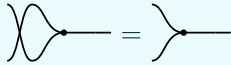
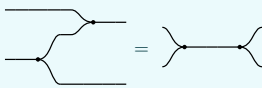
$$\begin{aligned} & \cdot f \dashv \cdot f^\circ \dashv / f^\circ \\ & f^\circ \cdot \dashv f \cdot \dashv f \backslash \end{aligned}$$

## §0.4. Frobenius

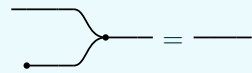
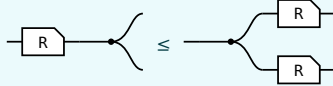
### Frobenius.



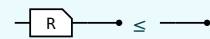
▷ associative



▷ commutative



◻ is its unit

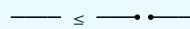
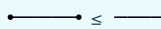
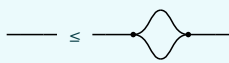
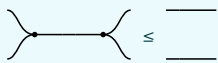


(0.4a)

Frobenius, one half — the other is its  $^{\circ}$

$R \triangleleft \leq \triangleleft (R \otimes R)$

$R \dashv \leq \dashv$



(0.4b)

▷ ◁ ≤ 1

1 ≤ ◁ ▷

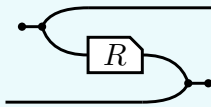
◻ ◻ ≤ 1

1 ≤ ◻ ◻

§0.5.  $\circ$  is a contravariant 2-functor

(0.5a)

$\circ$  is primitive, part of the data the first section lists. It turns both of  $R$ 's wires round, and the Frobenius structure DRAWS that — the picture and the formula below are  $R^\circ$ , not its definition:



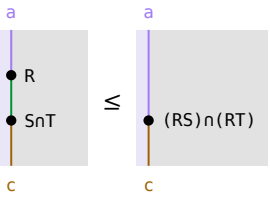
$$R^\circ = (\circ \text{--} \triangleleft \otimes 1) (1 \otimes R \otimes 1) (1 \otimes \triangleright \text{--} \circ)$$

where  $\circ \text{--} \triangleleft : \mathbb{I} \rightarrow a \otimes a$  opens a pair of wires out of nothing and  $\triangleright \text{--} \circ : b \otimes b \rightarrow \mathbb{I}$  closes one, so the input of  $R^\circ$  is where the output of  $R$  was. Taking that formula as the definition would now be circular:  $\triangleleft$  and  $\text{--} \circ$  are  $\triangleright^\circ$  and  $\circ \text{--}$ . The laws of  $\circ$  are that it is a contravariant 2-functor  $\circ : \mathcal{C}^\circ \rightarrow \mathcal{C}$ :

$$(i) \ 1^\circ = 1 \qquad (ii) \ (RS)^\circ = S^\circ R^\circ \qquad (iii) \ (R \otimes S)^\circ = R^\circ \otimes S^\circ \qquad (iv) \ R \leq S \text{ implies } R^\circ \leq S^\circ$$

§0.6. Meet and composition

(0.6a)

| semi-distributivity, and what supplies it                                                                                                               | picture                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| $R \ (S \cap T) \sqsubseteq RS \cap RT$ — the lax copy law. Equality exactly when $R$ is single valued: the Maps section's $F(R \cap S) = FR \cap FS$ . |  |

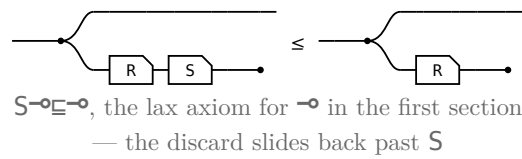
## §0.7. Domain and range

(0.7a)

|   |                                                                                                                                                                                                                                     |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | $\text{dom}(R) \triangleq 1 \cap (R \circ R^\circ)$                                                                                                                                                                                 |
| 2 | $\text{Coreflexive}(X) \implies \text{dom}(R) \subseteq X \iff R \subseteq X \times R$                                                                                                                                              |
| 3 | $\text{dom}(R \circ S) \subseteq \text{dom}(R)$                                                                                                                                                                                     |
| 4 | $\text{dom}(R \cap S) = 1 \cap (S \circ R^\circ)$                                                                                                                                                                                   |
| 5 | $R \text{ entire} \iff \text{Entire}(R) \triangleq \text{dom}(R) = 1$                                                                                                                                                               |
| 6 | $R \text{ simple} \iff \text{Simple}(R) \triangleq R^\circ \circ R \subseteq 1$                                                                                                                                                     |
| 7 | $R \text{ a map} \iff \text{Map}(R) \triangleq \text{Entire}(R) \wedge \text{Simple}(R)$                                                                                                                                            |
| 8 | $\text{Entire}(R) \wedge \text{Entire}(S) \implies \text{Entire}(R \circ S)$<br>$\text{Simple}(f) \wedge \text{Simple}(g) \implies \text{Simple}(f \circ g)$<br>$\text{Map}(f) \wedge \text{Map}(g) \implies \text{Map}(f \circ g)$ |
| 9 | $\text{Entire}(R \circ S) \implies \text{Entire}(R)$                                                                                                                                                                                |

## §0.8. Sliding the discard

(0.8a)



## §0.9. Division

|    |                                                                                                                                                                                                                 |  |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 1  | $X \in R/S \iff XS \in R$<br>$X$ is any $x$ -to- $y$ pairing; one that only pairs $x$ with a $y$ such that $x$ admires everyone $y$ hates lies inside $R/S$ , and $R/S$ is the largest such.                    |  |
| 2  | $X \in S \setminus R \iff SX \in R$<br>The mirror — divide on the left when $x$ comes first.                                                                                                                    |  |
| 3  | $(R/S)S \in R$<br>There is a $y$ such that $x$ admires everyone $y$ hates, and $p$ is one of the people $y$ hates — then $x$ admires $p$ too. Strict at $S=\emptyset$ : $R/S$ is everyone, $(R/S)S=\emptyset$ . |  |
| 4  | $S(S \setminus R) \in R$<br>The mirror.                                                                                                                                                                         |  |
| 5  | <b>associate:</b> $R/(S_1 S_2) = (R/S_2)/S_1$<br>A friend's enemy is two hops: divide by the far end first.                                                                                                     |  |
| 6  | $(S_1 S_2) \setminus R = S_2 \setminus (S_1 \setminus R)$<br>The mirror.                                                                                                                                        |  |
| 7  | <b>maps:</b> $f(R/S) = (fR)/S$<br>Rename $x$ before or after dividing — the licence to write $fR/S$ .                                                                                                           |  |
| 8  | $R/(fS) = (R/S)f^\circ$<br>Rename $y$ : a map leaves a denominator as $f^\circ$ outside the box.                                                                                                                |  |
| 9  | $(R/S)(S/W) \in R/W$<br>Someone who admires all of a hate-set that already covers everyone $z$ works for admires those people too.                                                                              |  |
| 10 | $1 \in R/R$<br>$R/R$ runs admirer to admirer: each admires everyone they admire. Strict: two people who each admire only $a$ and $b$ admire each other's idols too, and still stay two people.                  |  |
| 11 | $(R/R)(R/R) = R/R$<br>$R/R$ is the preorder <b>admires at least as much as</b> , and a preorder is idempotent. Freyd writes $\sqsubseteq$ ; with $1 \in R/R$ above it is an equality.                           |  |
| 12 | $(R/R)R = R$<br>Reaching $p$ through someone whose idols $x$ fully admires is reaching $p$ directly, since $x$ admires their own idols.                                                                         |  |
| 13 | $R/1 = R$<br>Dividing by $1$ : $p$ 's set is just $\{p\}$ , so admiring all of it is admiring $p$ .                                                                                                             |  |
| 14 | $R/(S_1 \cup S_2) = R/S_1 \cap R/S_2$<br>Admiring a combined hate-set is admiring each set in full.                                                                                                             |  |
| 15 | $S \setminus (R/W) = (S \setminus R)/W$<br>Which is why $S \setminus R/W$ needs no bracket.                                                                                                                     |  |

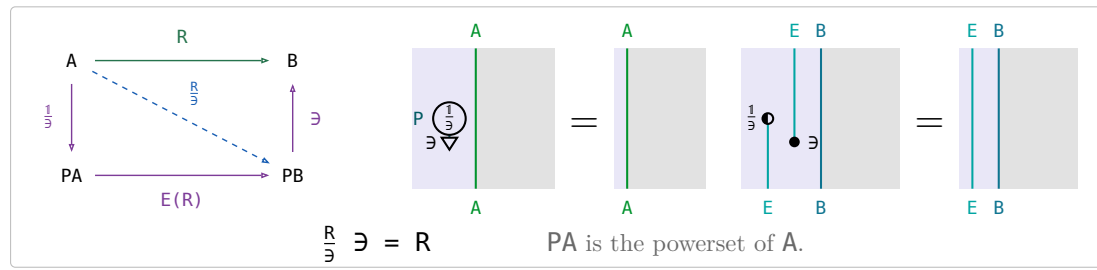
## §0.10. Power allegories

(0.10a)

|    | law                                                                                                                                                        | the reading                                                                                                                                                                                   |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | $\exists_R \square = R\square, \exists_R = \exists_R \square$                                                                                              | One $\exists$ per object, not per arrow.                                                                                                                                                      |
| 2  | $\exists$ is <b>thick</b>                                                                                                                                  | <b>Comprehension:</b> every $x$ has a set of exactly the people $x$ admires.                                                                                                                  |
| 3  | $\frac{R}{\exists} : A \rightarrow PB$ , for $R : A \rightarrow B$                                                                                         | convert a relation to a function. $a \frac{R}{\exists} = \{b \mid a R b\}$                                                                                                                    |
| 4  | $\text{Map}(\frac{R}{\exists})$                                                                                                                            |                                                                                                                                                                                               |
| 5  | $\frac{R}{\exists} \exists = R$                                                                                                                            | reading the chosen set back through $\exists$ returns the relation.                                                                                                                           |
| 6  | $\text{Simple}(F) \implies F \sqsubseteq \frac{F \exists}{\exists}$                                                                                        | A partial choice of sets is inside the total one.                                                                                                                                             |
| 7  | the <b>singleton map</b> is monic: $\frac{1}{\exists} (\frac{1}{\exists})^\circ \sqsubseteq 1$ two points with the same one-person set are the same point. | The one-person set.                                                                                                                                                                           |
| 8  | $\frac{\exists}{\exists} = 1$                                                                                                                              | Make the set of a set, then read it back one level down.                                                                                                                                      |
| 9  | <b>fusion:</b> $\text{Map}(f) \implies \frac{f R}{\exists} = f \frac{R}{\exists}$                                                                          | Naturality of the unit, $\text{Map}(f) \implies f \frac{1}{\exists} = \frac{1}{\exists} E(f)$ . renaming a point and then taking its one-person set is taking the set and renaming inside it. |
| 10 | $\text{Map}(f) \wedge 1\square = \exists\square \implies \frac{f}{\exists} = f \frac{1}{\exists}$                                                          | Rename first or take singletons first — the fusion row above at $R=1$ .                                                                                                                       |
| 11 | $R\square = \exists\square \implies R /_s S = \frac{R}{\exists} (\frac{S}{\exists})^\circ$                                                                 | $x$ and $y$ match when $R$ sends $x$ and $S$ sends $y$ to the same set.                                                                                                                       |
| 12 | $E(R) \triangleq \frac{\exists R}{\exists}$                                                                                                                | $E(R) : PA \rightarrow PB$ , image of a set of $A$                                                                                                                                            |
| 13 | $\frac{R}{\exists} = \frac{1}{\exists} E(R)$                                                                                                               | $\frac{1}{\exists} : x \mapsto \{x\}$                                                                                                                                                         |
| 14 | $\frac{S}{\exists} E(R) = \frac{S R}{\exists}$                                                                                                             | absorption — the monad's composition law, $\frac{S}{\exists} \diamond \frac{R}{\exists} = \frac{S R}{\exists}$ , $\S \mathcal{A} \cong \text{Kleisli}(E)$ in Relation Algebra                 |
| 15 | $E(R) = E(\frac{R}{\exists})$ union                                                                                                                        | the image of a set under $R$ is the union of the $R$ -sets of its points.                                                                                                                     |
| 16 | $\supseteq \triangleq \exists / \exists$                                                                                                                   | $xs \supseteq ys \iff \forall a. ys \exists a \rightarrow xs \exists a$                                                                                                                       |
| 17 | $\subseteq = \supseteq^\circ$                                                                                                                              | $xs \subseteq ys \iff ys \supseteq xs$                                                                                                                                                        |
| 18 | $T^\circ \frac{T}{\exists} \sqsubseteq E$                                                                                                                  | every $T$ -value of a point is a member of its $T$ -set: $\Lambda(T)$ is a map and $\Lambda(T)\exists = T$ .                                                                                  |

§0.11.  $i \dashv E$ : power transpose through singleton

(0.11a)

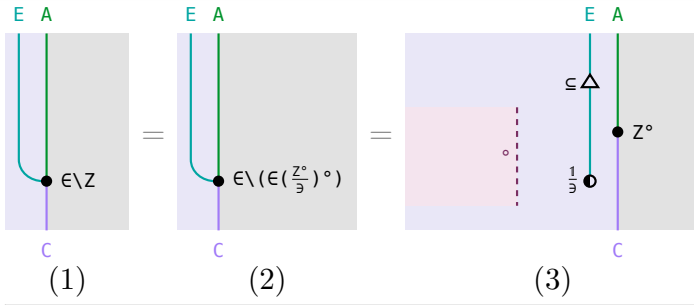


§0.12.  $\epsilon \setminus$  as a composite

$$\epsilon \setminus Z = \subseteq \left( \frac{Z^\circ}{\exists} \right)^\circ$$

(0.12a)

$xs$  is related to  $c$  by  $\epsilon \setminus Z$  exactly when  $xs \subseteq \Lambda(Z^\circ)(c)$ , the set of every  $a$  with  $a Z c$



|   |                                                                             |                                                                              |
|---|-----------------------------------------------------------------------------|------------------------------------------------------------------------------|
| 1 | $\epsilon \setminus Z$                                                      | $Z = \epsilon \wedge (Z^\circ)^\circ \text{ --- } \frac{R}{\exists} \ni = R$ |
| 2 | $= \epsilon \setminus (\epsilon \setminus (\frac{Z^\circ}{\exists})^\circ)$ | $X \setminus (Y f^\circ) = (X \setminus Y) f^\circ, f \text{ a map}$         |
| 3 | $= \subseteq \left( \frac{Z^\circ}{\exists} \right)^\circ$                  |                                                                              |